

(1) Fuzzification Phase

For the input variables, calculate their membership degrees in each linguistic set.

The **Consequent fuzzy set's** membership function is a monotonic function.

using the **Triangular Membership Function**:

$$\mu(x; a, b, c) = \begin{cases} 0, & x \leq a \text{ or } x \geq c \\ \frac{x - a}{b - a}, & a < x \leq b \\ \frac{c - x}{c - b}, & b < x < c \end{cases}$$

(2) Rule Aggregation

Use linguistic rules (If-Then) to describe the system:

| IF A AND B THEN C

(3) Inference Phas

Calculate the firing strength (α) for each rule:

$$\alpha_i = \min[\mu_A(x), \mu_B(y)]$$

(4) Output Mapping

The **THEN** part of each rule has a membership function that is monotonic.

The output (z_i) is obtained by reversing the membership function for the output:

$$\mu_{\text{output}}(z_i) = \alpha_i$$

That is:

$$z_i = \mu_{\text{output}}^{-1}(\alpha_i)$$

Thus, each rule in the Tsukamoto system gives a crisp output value.

(5) Defuzzification

The final output is the weighted average of all the rules:

$$z^* = \frac{\sum_i \alpha_i z_i}{\sum_i \alpha_i}$$